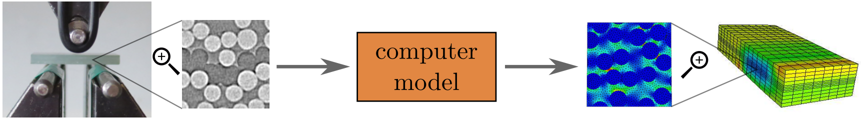
Materials and computer models – avenues for progress

The answer: combining a handful of experiments with extensive virtual testing

Multiscale modeling

- Model materials at very small and very large scales at the same time
- More knowledge $\Rightarrow$ smaller safety factors $\Rightarrow$ more efficient designs

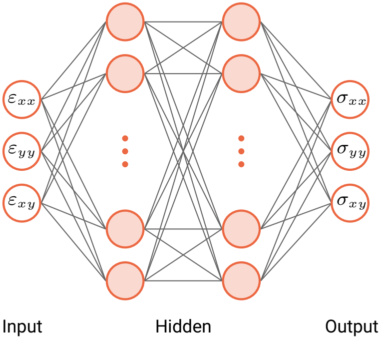


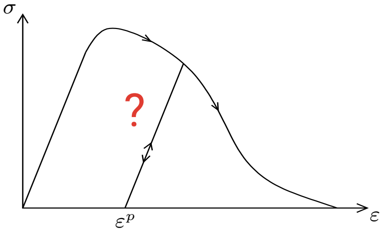
The challenge: multiscale modeling requires an immense amount of microscale computations

Neural network surrogates for multiscale

Neural nets are universal approximators, so they are a popular choice:

- Conventional networks can map strains to stresses but cannot handle path dependence





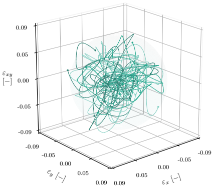
PRNNs versus conventional ML models

Okay, how about training with the challenging paths?

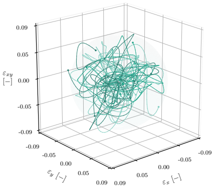
PRNN (36 curves)

RNN (576 curves)

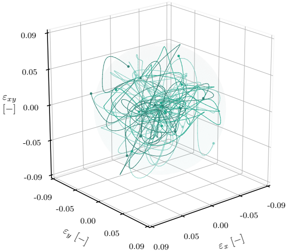
Train

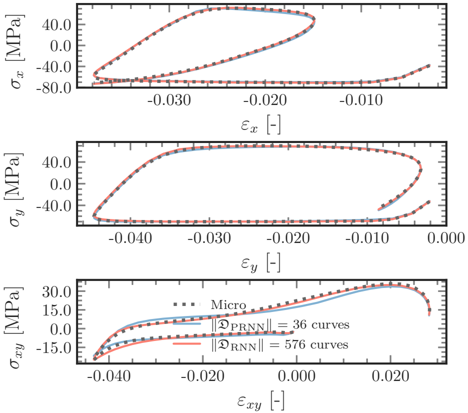


or



Test





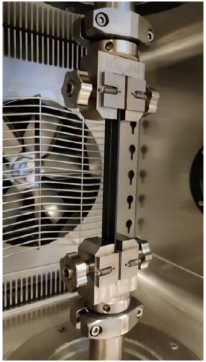
[Maia, Rocha, Kerfriden, van der Meer (2023), **CMAME**]

What we are now able to do: FE^2 versus experiments

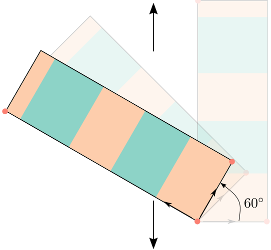
We also used it to confirm some experimental results that had been puzzling us:

- Uniaxial tension on off-axis carbon-PEEK coupons

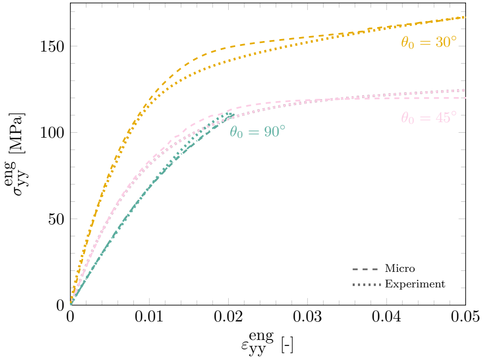
Uniaxial tension experiment



Off-axis angle



Engineering stress-strain



[Maia, Rocha, van der Meer (2025), **arXiv:2501:10193**]

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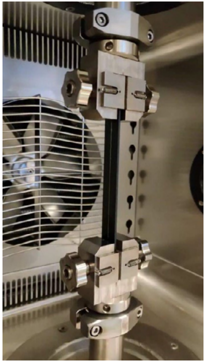
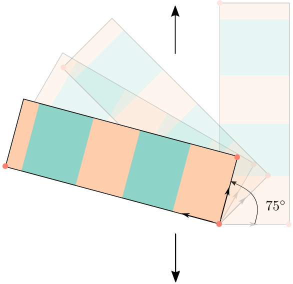


Off-axis angle



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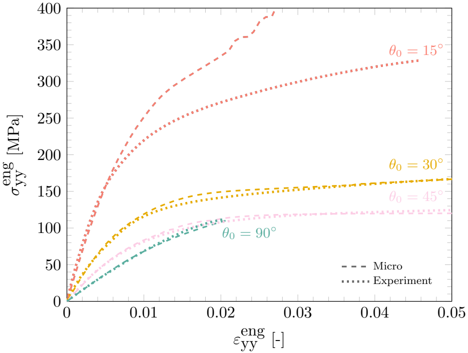

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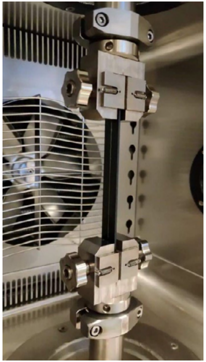
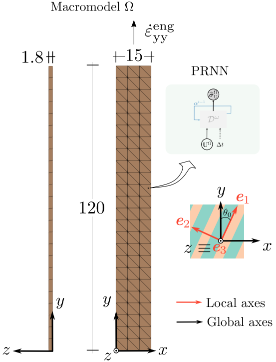


Multiscale model



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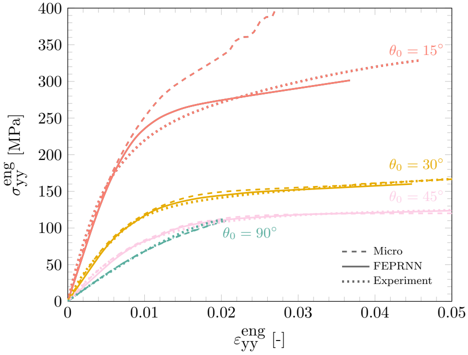
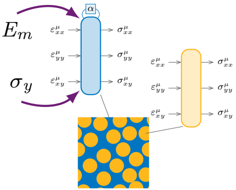


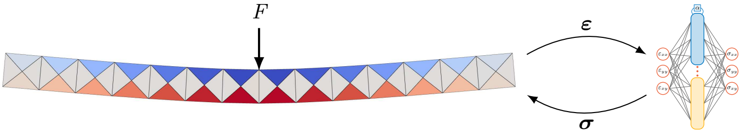
[Maia, Rocha, van der Meer (2025), **arXiv:2501:10193**]

What we are now able to do: Multiscale UQ

Three-point bending, coarse macro mesh:

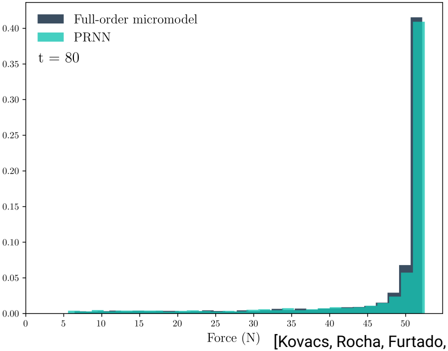
- Monte Carlo forward UQ, $N = 5000$ samples
- PRNN trained with a single (E_m, σ_y) pair
- Good agreement with full-order FE^2

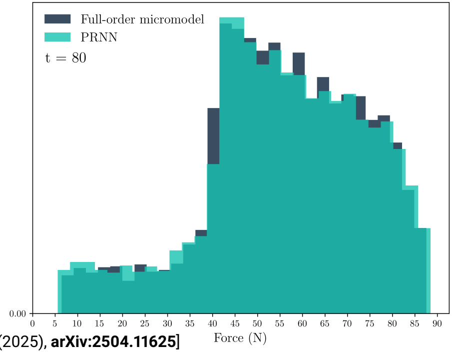




$$p(F) = \int p(F|E_m)p(E_m)\,\mathrm{d}E_m$$

$$p(F) = \int p(F|E_m,\sigma_y)p(E_m,\sigma_y)\,\mathrm{d}E_m\mathrm{d}\sigma_y$$



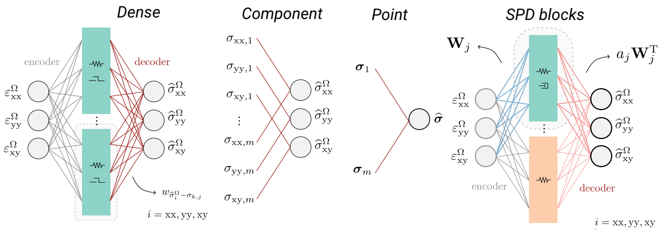


[Kovacs, Rocha, Furtado, Camanho, van der Meer (2025), **arXiv:2504.11625**]

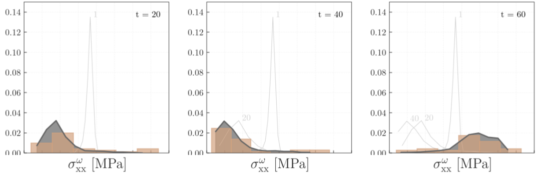
But wait, what about thermodynamics!?

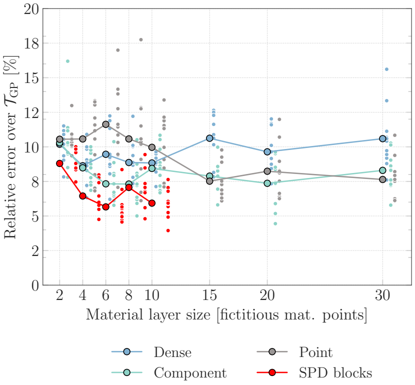
Architecture flexibility allows for a lot of experimentation:

- Sparse decoder connectivities improve interpretability and local stress distributions
- SPD encoder/decoders with shared weights guarantee thermodynamic consistency



Stress histograms for a random GP path (*Component* architecture)





[Maia, Rocha, Meyer, van der Meer (2025), **In preparation**]

Other recent works in the lab: Bayesian FEM

Assume a prior over the solution on a fine mesh, condition on shape functions from a coarse mesh:

Conventional FEM

Bayesian FEM

[Poot, Kerfriden, Rocha, van der Meer (2025), **arXiv:2506.02815**]

Thank you!

Feel free to check our website:

- Links to all of our code repositories
- Links to papers and preprints
- Slides from talks, including this one ;)

We are looking for interesting applications!